MILITARY SPECIFICATION

SIMULATOR, PROJECTILE, AIRBURST, LIQUID (SPAL): M11

This specification is approved for use within U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and *is* available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of liquid airburst projectile simulator.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-585 - Boxes, Wood, Wirebound
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-F-320 - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-P-704 - Pails, Metal: (Shipping, Steel, 1-12 Gallons)

MILITARY

MIL-P-223 - Powder, Black

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

AMSC N/A FSC 1370

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-810. - Environmental Test Methods
MIL-STD-1167 - Ammunition Data Card
MIL-STD-1168 - Ammunition Lot Numbering

2.1.2 Drawings and publications. The following drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

TDPL 15-12-248 - Simulator, Projectile, Airburst, Liquid: M11

PUBLICATIONS

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

EA-P-1172 - Polyethylene Glycol 200, Technical

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation.

ASTM STANDARDS

D638 - Tensile Properties of Plastics
D1238 - Flow Rates of Thermoplastics by Extrusion Plastometer
D1248 - Polyethylene Plastics Molding and Extrusion Materials
D1505 - Density of Plastics by the Density-Gradient Technique
D1693 - Environmental Stress-Cracking of Ethylene Plastics
D1928 - Preparation of Compression-Molded Polyethylene Test Sheets and Test Samples
D2240 - Rubber Property-Durometer Hardness

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(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3* REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on drawings in TDPL 15-12-248 and on subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the M11 SPAL shall conform to the specifications and drawings listed in TDPL 15-12-248 and on subsidiary drawings.

3.1.2.1 Molded plastic parts. The molded plastic parts (obturator, burster housing, housing cap, and bottles) shall be manufactured to conform to the following requirements:

(a) Color. Color shall be natural.

(b) Surface. The molded component surfaces shall be free of surface defects. Also, component surfaces shall be free of dirt, dust, grease, foreign matter, burns, part line and flash.

(c) Threads. All threaded sections of plastic parts shall have smooth clean threads free of surface molding defects.

(d) Markings. Each molded part shall have molded thereon the manufacturer's trademark, cavity number, and molding date marks as required by the applicable drawings or specifications and contract or purchase order. If part, but not all, of the markings listed can be shown on the molding, the order of preference shall be as listed in the preceding sentence. The trademark would receive first preference.

3.1.2.2 Obturator. The obturators, with 0.25 inch radius end dimension shall pass through a ring gage of 3.685 + .001 inches diameter by 1.25 ± .04 inches long when applying a force not exceeding 5 pounds (lb).

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3.1.2.3 Burster housing. Molded burster housings shall be molded from Phillips Corporation HMN 4550 high-density polyethylene-hexene copolymer (virgin material). No reground material is permitted. The following additional requirements shall apply:

(A) Physical properties.

(1) Raw material. The properties of raw material shall conform to the values of table I.

(2) Molded burster housing.

(a) Stress cracking. Twenty-four (24) molded burster housings shall be stored at each of the following four separate environmental conditions for 4 weeks: (1) hot, dry (160°F); (2) cold (-34°F); (3) hot, wet (113°F); (4) cyclic for 4 weeks. Molded burster housing shall show no evidence of stress cracking.

(B) Process controls for molded burster housings.

(1) Injection molding parameters. The following injection parameters for molding the burster housing shall be established during preproduction and submitted to the Government prior to first article testing. Data which substantiate the burster housings produced over the molding parameters range for preproduction samples, which meet the requirements specified in 3.1.2.3(A)(2)(a), shall be submitted.

(a) Injection press. Cite name, size, and number of molding machine, and identify mold.

(b) Melt temperature. Cite range (°F of material out of nozzle).

(c) Injection pressure time. (As fast as possible).

1 High pressure. Cite pounds per square inch gage (psig), minimum.

2 Hold pressure. Cite psig, minimum.

3 High pressure time. (Minimum required to fill mold). Cite seconds, maximum.

4 Hold pressure time. Cite seconds, maximum.

(d) Mold temperature. (Core and cavities designed to provide uniform temperature). Cite range, °F.

(e) Starve feed. (Ram must bottom out on each shot, no packing permitted). Cite control procedure.

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TABLE I. High density polyethylene hexene (copolymer)

Property	Requirements	Test method	No. of samples
1. *Density, gm/cm^3	0.941-0.949	ASTM D1505, method B	3
2. Melt index gm/10 min	4.0-6.0	ASTM D1238, procedure A, condition E	2
3. *Tensile strength psi 2 in/min	3,000 minimum	ASTM D638, type IV specimens	5
4. *Ultimate elongation % @ 2 in/min	300 minimum	ASTM D638, type IV specimens	5
5. *Hardness Shore D	52-60	ASTM D2240	2
6. ESCR, condition A, hours, F_{50} full strength Igepal CO.630 as reagent	24 minimum	ASTM D1693, condition A	10
7. Color	Natural	Visual	Bulk

*Test specimens shall be prepared in accordance with Procedure C of ASTM D1928 as it would be applied to material conforming to type III, class A (natural color) of ASTM D1248.

NOTE 1: The number of samples specified in the above table shall be selected randomly from each raw material lot and be subjected to the above inspections.

NOTE 2: No mold release shall be permitted.

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(f) Release agent. None.

(g) Total cycle. Cite time, minimum seconds.

(h) Weight of housing. Cite maximum, grams.

(2) Maintenance of process controls throughout production. A production plan to assure that process control (see 3.1.2.3(B)(1)) shall not be changed without approval by the Government.

(C) Process certification for molded burster housings. Production of molded burster housings shall conform to controls established during preproduction. Actual data shall be recorded hourly continuously during production for all controls specified in this specification. This certified data shall be submitted to the Government and the values shall fall within the ranges approved. Whenever production equipment, process controls, or production plans require change, new production samples shall be submitted for approval to the Government.

3.2 Manufacture and assembly. The M11 SPAL shall be produced and assembled in accordance with Drawing 15-12-248.

3.3 Leakage for housing cap assembly affixed to housing. The housing cap assembly (Drawing 15-12-243), when affixed to the housing (without explosive train) (Drawing 15-12-236), shall not show any evidence of leakage when tested as specified in 4.4.4.4.

3.4 Functioning.

3.4.1 Burst height. The completely assembled M11 SPAL shall have a minimum height of burst of 15 meters when tested as specified in 4.4.4.1.

3.5 Fracture of housing cap. The housing cap shall fracture or rupture when subjected to a load of *135.0 + 2.0* lb applied for a maximum of 5 seconds when tested as specified in 4.4.4.2.

3.6 Preproduction. Prior to the start of regular production, preproduction samples shall be produced in accordance with this specification for examinations and tests.

3.7 Moisture content of black powder (Drawing 15-12-238). The moisture content of black powder shall not exceed 0.70 percent at the time and place of loading when tested as specified in 4.4.4.3.

3.8 Workmanship. The assembly and components shall be free of burrs, sharp edges, cracks, dirt, grease, oil, and flash.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in *the* contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government *for* acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.2 and section 5, for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspection. The inspection requirements are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Preproduction lot. A preproduction lot of components suitable to assemble and pack a minimum of 750 M11 SPAL shall be produced using the same methods, materials, equipment, and processes as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination and nondestructive tests. A random sample of 15 each components, subassemblies, assemblies and packaging along with 15 packed M11 SPAL shall be examined 100 percent to all drawings dimensions/requirements and this specification. The sample for moisture content of black powder shall be conducted in accordance with MIL-P-223.

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4.3.2.2 For destructive tests. The raw material of burster housings shall be tested in accordance with table I. The 100 housing caps shall be tested in accordance with 4.4.4.2. Twenty-four burster housings from each cavity, minimum of *96* burster housings, shall be tested in accordance with 3.1.2.3(2)(a). The burst height for SO assembled (10 sets of five) M11 SPAL shall be tested in accordance with 4.4.4.1.

4.3.3 Acceptance/rejection criteria. The preproduction samples shall meet all the requirements for examinations and tests listed in the drawings and this specification to be acceptable. The acceptance number is zero. The process control values submitted shall be acceptable only if the burster housing produced thereby meet the requirements as verified by the actual test data designated in this specification. The procedure for process control certification shall likewise be acceptable to the Government. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of like components or subassemblies (including explosive material) of one interfix from one manufacturer, from the same materials, and under essentially the same manufacturing conditions. Each lot shall be identified and controlled in accordance with MIL-STD-1168. Data cards shall be prepared for each lot in accordance with MIL-STD-1167.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105. Sampling for the 5-gallon steel drum shall be conducted in accordance with MIL-STD-105 using an AQL of 0.65 percent defective.

4.4.2.2 For destructive tests. Sampling for the raw material of the burster housing shall be conducted in accordance with table I with an acceptance number of o. Sampling for housing caps and burst height shall be conducted in accordance with MIL-STD-105, level S-2 using an AQL of 0.65 percent defective. Moisture content of black powder shall be conducted in accordance with MIL-P-223.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. Samples shall be examined and tested in accordance with the classification of defects and MIL-STD-105. The 5-gallon steel drum shall be tested in accordance with 4.4.4.5 using an AQL of 0.65 percent defective.

4.4.3.2 For destructive tests. The raw material of the burster housing shall be tested in accordance with table I. Housing caps shall be tested in accordance with 4.4.4.2. The burst height for the completely assembled M11 SPAL shall be tested in accordance with 4.4.4.1. Moisture content of black powder shall be tested in accordance with the provisions of MIL-P-223.

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4.4.3.3 For critical defects. Each item in the lot shall be tested for critical characteristics listed in the classification of defects.

4.4.3.4 Classification of defects.

(a) Case, detonator (Drawing 15-12-336).

Categories		Defects
Critical:		None defined
Major:	AQL	1.0 percent defective
101		Case length incorrect
102		Inside diameter incorrect
103		Outside diameter incorrect
104		Bottom thickness incorrect

(b) Detonator, assembly (Drawing 15-12-335).

Categories		Defects
Critical:		None defined
Major:	AQL	1.0 percent defective
101		Depth to ignition charge (.562 + .062) incorrect

(c) Explosive delay train (Drawing 15-12-242).

Categories		Defects
Critical:		None defined
Major:	AQL	1.0 percent defective
101		Component missing or incorrectly assembled
102		Tape wrinkled
103		Detonator sized incorrectly
104		Cord length incorrect
105		Cord ends not sealed

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(d) Housing cap (Drawing 15-12-237).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Outside diameter incorrect (1.245)
102	Inside diameter incorrect (.870)
103	Rad (male bead) missing or dimensions incorrect
104	Mold plastic part incorrect (3.1.2.1)
105	Gates or sprue not trimmed flush
106	Workmanship (3.8)

(e) Housing cap assembly (Drawing 15-12-243).

Categories	Defects
Critical:	
1	Electrical match wires not shunted
Major:	AQL 1.0 percent defective
101	Epoxy missing
102	Workmanship (3.8)

(f) Housing (Drawing 15-12-236).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Evidence of stress cracking [3.1.2.3(A)(2)(a)]
102	Rad (female head) missing or dimensions incorrect
103	Diameter (.941 ± .006) incorrect
104	Diameter (1.235) incorrect
105	Wall thickness (.075 rein) incorrect
106	Diameter (.285 ± .005) incorrect
107	Gates or sprue not trimmed flush
108	Mold plastic part incorrect (3.1.2.1)
109	Length (7.80 max) incorrect
110	Workmanship (3.8)

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(g) Explosive housing assembly (Drawing 15-12-238).

Categories	Defects	Acceptance standards
Critical:		
1	Electric match wires not shunted	
Major:	AQL 1.0 percent defective	
101	Component missing or incorrectly assembled	
102	Label missing or wrinkled	
103	Sealant not applied to housing cap assembly	
104	Leakage of sealed housing cap assembly affixed to housing	4.4.4.4
105	Workmanship (3.8)	

(h) Obturator screw closure (Drawing 15-12-239).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Diameter (3.680) incorrect
102	Diameters (1.375 rein) incorrect
103	Mold plastic *part* incorrect (3.1.2.1)
104	Gates or spruce not trimmed flush
105	Ring gage force excessive (3.1.2.2)
106	Diameter (.785) incorrect
107	Threads incorrect
108	Workmanship (3.8)

(i) Bottle assembly (Drawing 15-12-244).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Markings missing, incorrect, or illegible
102	Label applied incorrectly
103	Workmanship (3.8)

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(j) Bottle/closure (Drawing C15-12-240).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Min inside length (7.900)
102	Mold plastic part incorrect (3.1.2.1)
103	Workmanship (3.8)

(k) Label (Drawing 15-12-254).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Markings missing, incorrect, or illegible
102	Color incorrect
103	Workmanship (3.8)

(1) Packaging (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Component missing
102	Preservation incorrect (5.1)
103	Packing incorrect (5.2)
104	Marking incorrect (5.3)

4.4.4 Test methods and procedures. Warning: This specification covers sampling and testing of explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations, and procedures must be followed in handling and processing these materials.

4.4.4.1 Burst height. The completely assembled M11 SPAL shall have the specified burst height when fired from a launcher, projectile, airburst, liquid (SPAL): M267 (Drawing 13-12-145) in accordance with Operator's and Organizational Maintenance Manual DEP TM 3-6920-100-12 & P (see 6.3).

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4.4.4.2 Fracture of housing cap. The 0.870 diameter surface of the housing cap shall be subjected to a uniformly applied force as required. Fracture or rupture of the housing cap shall be within 5 seconds maximum when tested at 25″ + 2°c.

4.4.4.3 Moisture content. The moisture content of the black powder shall be tested in accordance with MIL-P-223 just prior to assembly of the explosive housing.

4.4.4.4 Leakage for housing cap assembly affixed to housing. The housing cap assembly, when affixed to the housing, shall be tested in accordance with procedure I of MIL-STD-810.

4.4.4.5 Pull test on 5-gallon steel drum. When placed in a suitable test fixture the empty 5-gallon steel drum shall be held by the handle and subjected to a direct pull of 100 lb. The handle and drum shall be observed for damage. Any separation of the welds shall be cause for rejection.

4.4.5 Acceptance/rejection criteria. Failure of the sample to meet the specified examinations and tests shall be cause to reject the lot from which the sample was drawn and to cease those operations causing failure until corrective action has been accomplished and approved by the Government. Failure of certified process control data to meet approved values shall be cause to reject the lot, realign the process, and begin a new lot of molded burster housings.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or B as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Explosive burster housing assembly (Drawing 15-12-238). Fifteen burster housing assemblies shall be unit packed in accordance with Drawings 15-12-256 and 15-12-253, as shown.

5.1.1.2 Obturator (Drawing 15-12-239) and bottles (Drawing 15-12-244). Fifteen bottles (empty) and fifteen obturators shall be unit packed together in accordance with Drawing 15-12-250, as shown.

5.1.1.3 Faucet (Drawing 15-12-339). The faucet shall be unit packed in accordance with Drawing 15-12-246.

5.1.1.4 Polyethylene glycol (PEG) *200* (5-gallon quantity). A quantity of 5 U.S. gallons of PEG 200 (EA-P-1172) shall be unit packed in a nominal 5-gal steel drum. The drum shall conform to type 1, class 2, of PPP-P-704 and shall have an interior coating consisting of a minimum of one coat of epoxy-phenolic baking enamel. The ‘*D’* type handle shall be secured to the head of the drum insuring that there shall be no evidence of damage to the head of the drum or

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handle, or separation of the handle from the drum when tested as specified in 4.4.4.5. Also, the drum shall be furnished with two nominal 3/4-inch National Pipe Thread (N.P.T.) metal closure assemblies, each diametrically opposed in the top head of drum, and positioned with respect *to* the chime and handle so as to be easily closed and opened by hand tools. The closures shall consist of standard industrial plug and flange assemblies, each fitted with minimum 1/16-inch thick neoprene gaskets of durometer hardness range of 40 to 65. The plug shall be designed to be removable *by* using a standard drum plug wrench or a nominal 1/2 inch wide screwdriver. The plug shall be closed *to* a torque within a range as specified by the drum manufacturer. There shall be no evidence of leakage in the filled drum.

5.1.2 Level B.

5.1.2.1 Explosive burster housing assembly (Drawing 15-12-238). Fifteen burster housing assemblies shall be unit packed as specified in 5.1.1.1.

5.1.2.2 Obturator (Drawing 15-12-239) and bottles (Drawing 15-12-244). Fifteen bottles (empty) and obturators shall be unit packed as specified in 5.1.1.2 except that the fiberboard box, fiberboard interlocking partitions, and fiberboard pads shall conform to RSC, type CF, class domestic, variety SW, grade 175 or 200 of PPP-B-636 and type CF, class domestic, variety SW, grade 175 or 200 of PPP-F-320, respectively.

5.1.2.3 Faucet (Drawing 15-12-339). The faucet shall be unit packed as specified in 5.1.1.3.

5.1.2.1 Polyethylene glycol (PEG) 200 (S-gallon quantity). A quantity of 5 U.S. gallons of PEG 200 shall be unit packed as specified in 5.1.1.4.

5.2 Packing. Packing shall be level A or B as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Explosive burster housing assembly (Drawing .15-12-238), obturator (Drawing 15-12-239), bottles (Drawing 15-12-244), and faucet (Drawing 15-12-339). The explosive burster housing assembly, obturator, bottles, and faucet, unit packed as specified in 5.1.1, shall be packed together in accordance with Drawing 15-12-246.

5.2.2 Level B.

5.2.2.1 Explosive burster housing assembly (Drawing 15-12-238), obturator (Drawing 15-12-239), bottles (Drawing 15-12-244), and faucet (Drawing 15-12-339). *The* explosive burster housing assembly, obturator, bottles, and faucet, unit packed as specified in 5.1.2, shall be packed together in accordance with Drawing D15-12-246, except that the shipping container shall be class 1, style 2 of PPP-B-585.

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5.2.2.2 Polyethylene glycol 200 (5-gallon quantity). Polyethylene glycol 200, unit packed as specified in 5.1.1.4, shall require no further protection for shipment aside from unitizing the 5-gallon quantity.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.2), marking shall be accomplished in accordance with MIL-STD-129 and Drawings 15-12-253, 15-12-251, and 15-12-334, as applicable.

6. NOTES

6.1 Intended use. The M11 SPAL is intended for use in tactical training exercises.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction.

(1) Time allowed for cantractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for Government to notify the contractor whether or not to proceed with production.

(c) Level of presentation required (A or B) (see 5.1).
(d) Level of packing required (A or B) (see *5.2).*
(e) Marking required (see 5.3).

6.3 Warnings and precautionary notes. The warnings and precautionary notes in DEP TM 3-6920-100-12 & P should be observed.

6.4 Subject term (key word) listing.

Polyethylene glycol 200
Simulator
SPAL

6.5 Supersession data. This specification supersedes and includes the requirements of Chemical Systems Laboratory Purchase Description EA-S-1266, dated 30 June 1983.

Custodian:

Army - EA

Preparing activity:

Army - EA

project No. *1370-A206*

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-51528(EA)	SIMULATOR, PROJECTILE, AIRBURST, LIQUID (SPAL): M11

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR
	☐ USER
b. ADDRESS (Street, City, State, ZIP Code)	☐ MANUFACTURER
	☐ OTHER (Specify): _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

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